

DATASET

Information	
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Affiliation	Department of Structural Engineering and Building Materials, Ghent University
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1 PREAMBLE

This dataset presents the early research outcomes from the initial experimental test program under the CARBCOMN project. The program was designed as the first iteration of an optimization protocol aimed at developing extrudable mixes based on slag-only formulations, i.e 100% secondary constituents that solely rely on carbonation curing for full strength development. The primary goal was to identify parameter ranges that enable reliable extrusion while maintaining lower water-to-binder ratios to facilitate a suitable microstructure for CO₂ diffusion and desirable mechanical performance.

The dataset provides a comprehensive record of the early findings and brings together information from several experimental domains. Specifically, it contains:

- **Mix design data**, documenting the proportions of raw materials and admixtures used in the slag-based systems.
- **Mechanical performance data**, including raw compression and tensile strength test results at different stages.
- **Micro-CT data**, consisting of initial representative specimens to assess internal porosity and grain distribution.
- **Rheological measurements**, provided both as raw data and selected figures, illustrating time-dependent flow behaviour and yield stress evolution of optimized formulations.
- **Preliminary thermogravimetric analysis (TGA)**, of selected mixes, providing insights into binder decomposition, carbonation reaction and stability.
- **Experimental setup documentation**, including photographs and schematics of the testing arrangements, specimen geometries and conditions.

Collectively, these files capture the first cycle of optimization for printable slag-based formulations. While preliminary, this dataset establishes baseline relationships between mixture design, printability, microstructural features and mechanical performance. It therefore serves as both a record of the initial test program and a reference point for subsequent optimization cycles.

2 FILES AND NOMENCLATURE

This section provides detailed description of the data files contained in the dataset, including their structure and naming conventions. For each excel file, the corresponding sheets are specified, along with the column input parameters, and units. This section also clarifies the annotations and acronyms applied across datasets. Both raw and processed data are documented, with explanation of how they are distinguished and linked. This ensures consistency, transparency and ease of use when accessing and interpreting the dataset.

2.1 08.2025_mixes_strength-results

- **SHEETS**
 1. **AC_IFSTTAR**: Contains data on water absorption capacity (AC) of secondary aggregates (Stinox) in accordance with the No. 78 IFSTTAR method, after 120 minutes of soaking.
 2. **Mix Design&Proportioning**: Provides mix proportions of binder (Carbinox), aggregates, water used in the mix (after corresponding correction due to AC of aggregates) and admixtures.
 3. **Strength_3%CO2_full prog.**: Flexural and compressive strength test results from the initial test program with mix design defined in sheet 2. The results are obtained after 7 days of

carbonation curing of small-scale specimens under 3% CO₂ concentration, 20° C temperature, and 60% relative humidity.

4. **Strength_20%CO2_7 d.:** Flexural and compressive strength test results from the selected reference mix M-0.5_0.25 (proportions defined in sheet 2). The results are obtained after 7 days of carbonation curing of small-scale specimens under 20% CO₂ concentration, 20° C temperature, and an uncontrolled relative humidity.
5. **Strength_20%CO2_1 d.:** Flexural and compressive strength test results from the selected reference mix M-0.5_0.25 (proportions defined in sheet 2). The results are obtained after 1 day of carbonation curing of small-scale specimens under 3% CO₂ concentration, followed by 1 day under 20% CO₂ concentration, both at 20° C temperature, and 60% relative humidity.
6. **Strength_20%CO2_2 d.:** Flexural and compressive strength test results from the selected reference mix M-0.5_0.25 (proportions defined in sheet 2). The results are obtained after 1 day of carbonation curing of small-scale specimens under 3% CO₂ concentration, followed by 2 days under 20% CO₂ concentration, both at 20° C temperature, and 60% relative humidity.
7. **Block Strength_3% CO2_7d.:** Flexural and compressive strength results of test specimens extracted from the 3D-printed trial block (fig. 2), batched following the selected reference mix M-0.5_0.25 (proportions defined in sheet 2). The results are obtained after 7 days of carbonation curing of the trial block under 3% CO₂ concentration, 20° C temperature, and an uncontrolled relative humidity.

• GENERAL & RAW DATA COLUMNS

Time (m), M_{SSD}, M_{OD} as recoded and defined in respective sheet

W_i and H_i; single-point overall width and thickness (height) respectively of standard test specimens, measured at 3 different points across the specimen in millimetres.

Cr've load (N); Compressive force at which the specimen failed the compression, measured in Newton (N)

Rupture load (N); Force at which the specimen failed in flexure, measured in Newton (N)

Direction; Direction of applied load with respect to the specimen geometry and sign convention shown in in fig. 1

• PROCESSED DATA COLUMNS

AC (%); percentage amount of water absorbed by the secondary aggregates over the soaking period measured relative to the oven dry mass of the aggregates.

W, H; average specimen width and thickness (height) of W_{i-n} and H_{i-n} respectively.

Cr've Strength (MPa); compressive strength of the measured specimen.

Avg. Cr've Strength (MPa); average compressive strength of total measured specimens.

Flex. Tensile Strength (MPa); flexural strength of the measured specimen.

Avg. Flex. Strength (MPa); average flexural strength of total measured specimens.

Stdev.; standard deviation of measured parameter.

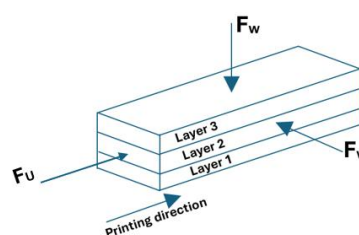


Figure 1 - Adopted loading sign convention

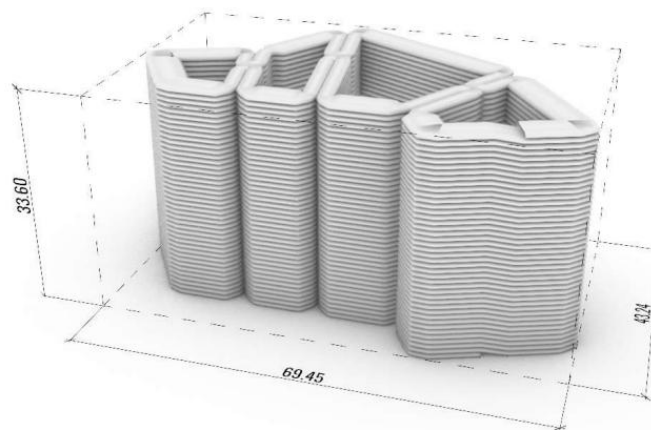


Figure 2 – Geometry of 3D-printed trial block (dimensions are in cm)

2.2 08.2025_Rheology tests

- **SHEETS**

1. **Flow table tests;** Contains data on spread diameter of different mixes after 25 table drops, following the ASTM 1437-20.
2. **Slow penetration_Normal Force;** Provides recorded normal force required to achieve a constant penetration of 5 μ m/s.
3. **Slow penetration_yield stress;** processed date of calculated yield stress values.
4. **Queue flow_Stress-strain;** Data from squeeze flow tests, enlisting recorded compression force, deformations and average stress-strain processed output.

- **GENERAL & RAW DATA COLUMNS**

Time (m); time lab between two successive tests

D₁ & D₂; Table spread diameter measured in perpendicular directions **1** and **2** respectively, in millimetres.

NF₁(N) & NF₂(N); Normal penetration force (in Newtons, N) from slow penetration tests. 1 and 2 represent different batches of the same mix.

Force1 (N) & Forece2 (N); Compression force (in Newtons, N) from squeeze flow tests. 1 and 2 represent different batches of the same mix.

Def1(mm) & Def2 (mm); Specimen deformation (in millimetres) from squeeze flow tests. 1 and 2 represent different batches of the same mix.

- **PROCESSED DATA COLUMNS**

D_{T1} & D_{T2}; Average spread diameter of **D₁** and **D₂** in millimetres.

D_{avg}; Average spread diameter of **D_{T1}** & **D_{T2}**, in millimetres.

NF₀(N) & NF₃₀(N); Average of **NF₁(N) & NF₂(N)**, respectively measured immediately (T = 0) and 30 minutes (T = 30) after mixing.

σ_0 (Pa) & σ_{30} (Pa); Average yield stress values (in Pascals) derived from **NF₀(N) & NF₃₀(N)**.

Stress (kPa); Stress in kilopascals, derived from the average of **Force1 (N) & Forece2 (N)** and sample dimensions.

Avg. strain; Average strain values derived from **Def1(mm) & Def2 (mm)** and specimen dimensions.

2.3 08.2025_TGA

- **SHEETS**
 1. **TGA_Raw**; Contains mass (in kilograms) data with respective temperatures (in degrees Celsius)
 2. **TGA_mass-loss**; Provides data on percentage mass loss with temperature
- **GENERAL & RAW DATA COLUMNS**

Temp (°C); Temperature (in degrees Celsius) to which the specimen power was heated

1_d (mg) & 2_d (mg); Mass (in milligrams) of M-0.5_0.25 powder at a specific temperature. The powder is extracted after 1 day of carbonation curing of small-scale specimens under 3% CO₂ concentration, followed by 1 day and 2 days respectively, under 20% CO₂ concentration, 20° C temperature, and 60% relative humidity.

Cx (mg); Mass (in milligrams) of uncarbonated binder (Carbinox) at a specific temperature.
- **PROCESSED DATA COLUMNS**

1_d (%), 2_d (%) % Cx (%); Percentage mass loss with increase in temperature derived from **1_d (mg)**, **2_d (mg)** and **Cx (mg)**

2.4 08.2025_Micro-CT data

- **SHEETS**
 1. **Equivalent_diameter**; Contains diameter of pores of CT-scanned samples cored from M-1.0_0.35 & M-0.7_0.25 small-scale prints, cured under 3% CO₂ concentration, 20° C temperature, and 60% relative humidity for 7 days.
 2. **Pore_volume_fraction**; Provides data on porosity distribution in different sample (described in sheet 1) axes.
- **PROCESSED DATA COLUMNS**

EQ_M-1.0_0.35 & EQ_M-1.0_0.35; Equivalent diameter (in micrometres) of each pore/void in cored samples of M-1.0_0.35 and M-0.7_0.35. Samples are CT-scanned using coreTom scanner and data processing is carried out using AVISO software.

PV-M-1.0_0.35 & PV_M-1.0_0.35; Ratio of volume of pores to volume of 2D slice in the considered sample axis Z and Y.

3 ADDITIONAL INFORMATION

Table 1 - Summary of the mixing protocol

Step	Constituent (s)	Added after (time, sec)	mixed for (time, s)	Rotation/revolution Speed (rpm)
1	Sand & binder	0	30	140/62.5
2	90-95% mixing water	30	30	140/62.5
3	5-10% mixing water + SP	60	30	285/125
4	VMA	90	60	285/125

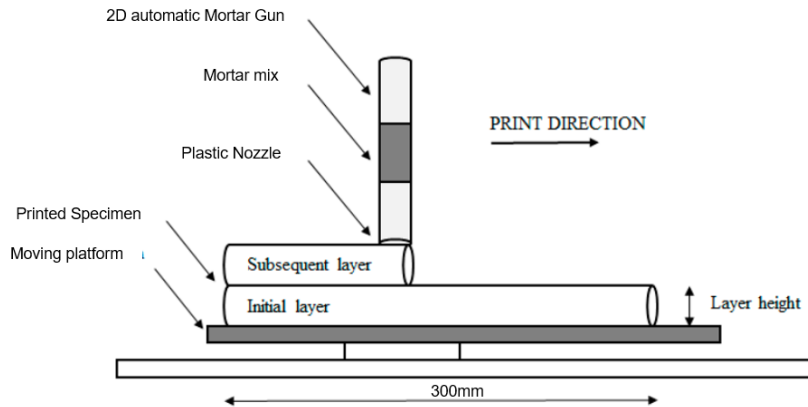


Figure 3 - 2D mortar-gun used for extrusion tests and printing of small-scale samples

Table 2 - Employed tests and standards

Property/Test	Purpose	Standard
Aggregate gradation	Determine Particle size distribution and gradation analysis of sands	EN 933-1
Properties of aggregates	Determine densities and water absorption of aggregates.	EN 1097-6 (Densities) No. 78 IFSTTAR (Water absorption)
Flow	Assess flow table values for conformity to extrudability requirements.	ASTM 1437-20
Carbonation	Assess depth of carbonation of carbon-cured concrete samples.	RILEM CPC-18
Compressive strength	Measure hardened-state compressive strength of printed or cored concrete samples.	EN 12390-3 EN 12504-1(Cored)
Flexural tensile strength	Measure flexural strength of printed or cored concrete samples.	EN 12390-5

Mix designation	a/b ratio	w/b ratio	Mass Per m ³ of concrete					
			Carbinox (kg)	Stinox (kg)	Effective Water, W (kg)	Total Water, W _T (kg)	SP (kg)	VMA (kg)
M-0.5_0.25	0.5	0.25	1060	530	265	282	8.5	0.6
M-0.5_0.30	0.5	0.30	1007	504	302	318	2.5	0.6
M-0.5_0.35	0.5	0.35	959	479	336	350	0.0	0.7
M-0.7_0.25	0.7	0.25	942	659	235	256	9.4	0.8
M-0.7_0.30	0.7	0.30	899	630	270	289	4.5	0.7
M-0.7_0.35	0.7	0.35	861	602	301	320	0.0	0.4
M-1.0_0.30	1.0	0.30	775	775	233	257	7.8	0.7
M-1.0_0.35	1.0	0.35	746	746	261	284	3.7	0.7

APPENDIX

A.1 Mixes in the initial test program

W_T – total amount of water added to the mix after correction due to water absorption of Stinox.